

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING AI & ML

LABORATORY OBSERVATION RECORD

STUDENT UTILITY PROGRAM IN JAVA

1. TITLE OF THE EXPERIMENT

Implementation of a Menu-Driven Student Utility Program Using Conditional Statements, Looping Statements and User-Defined Methods in Java

2. AIM

To design and implement a menu-driven Java program that demonstrates student information processing, conditional statements, looping statements, and reusable utility methods.

3. ALGORITHM

Step 1: Start the program and create a Scanner object.

Step 2: Display the Student Utility main menu.

Step 3: Read the user's choice.

Step 4: Use switch-case to select the required operation.

Step 5: For Student Information, read name, roll number and three marks; calculate total and percentage and display the result.

Step 6: For Conditional Statements, perform even/odd checking, largest-of-three calculation, grade determination, or day-of-week selection.

Step 7: For Looping Statements, generate a multiplication table, display 1 to N, calculate the sum of N natural numbers, or generate Fibonacci terms.

Step 8: For Utility Methods, call factorial(), isPrime(), maxOfTwo(), or areaOfCircle().

Step 9: Repeat the menu until choice becomes 0.

Step 10: Close Scanner and terminate the program.

4. EXECUTION STEPS

Step 1: Compile the StudentUtility Java program.

Step 2: Run the program and observe the main menu.

Step 3: Select option 1 and enter sample student details.

Step 4: Select option 2 and execute each conditional operation.

Step 5: Select option 3 and execute each looping operation.

Step 6: Select option 4 and execute each utility method.

Step 7: Select 0 to exit the program.

Step 8: Compare the actual output with the expected output.

5. SOURCE CODE

```
package task;
import java.util.Scanner;

public class StudentUtility {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int choice;
        do {
            System.out.println("StudentUtility program");
            System.out.println("1.Student Information");
            System.out.println("2.Conditional Statements");
            System.out.println("3.Looping Statements");
            System.out.println("4.Utility methods");
            System.out.println("5.Exit");
            System.out.println("Enter your choice:");
            choice=sc.nextInt();
            switch(choice) {
                case 1:
                    System.out.println("Enter Name:");
                    sc.nextLine();
                    String name=sc.nextLine();
                    System.out.println("Enter the Roll Number:");
                    int roll =sc.nextInt();
                    System.out.println("Enter Marks in 3 subjects");
                    int m1=sc.nextInt(),m2=sc.nextInt(),m3=sc.nextInt();
                    int total=m1+m2+m3;
                    double percentage =(total/3)*100;
                    System.out.println("\n--- Student Result ---");
                    System.out.println("Name: " + name);
                    System.out.println("Roll No: " + roll);
                    System.out.println("Total Marks: " + total);
                    System.out.println("Percentage: " + percentage + "%");
                    break;

                case 2:
                    System.out.println("\n Conditional Statement");
                    System.out.println("1. Even or Odd");
                    System.out.println("2. Largest of Three Numbers");
                    System.out.println("3. Grade based on Percentage");
                    System.out.println("4. Day of the Week");
                    int condChoice = sc.nextInt();

                    switch (condChoice) {
                        case 1:
                            System.out.print("Enter a number: ");
                            int num = sc.nextInt();
                            System.out.println(num % 2 == 0 ? "Even" : "Odd");
                            break;
                        case 2:
                            System.out.print("Enter three numbers: ");
                            int a = sc.nextInt(), b = sc.nextInt(), c = sc.nextInt();
                            int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
                            System.out.println("Largest: " + largest);
                            break;
                        case 3:
                            System.out.print("Enter percentage: ");
                            double per = sc.nextDouble();
                            if (per >= 90) System.out.println("Grade: A+");
                            else if (per >= 75) System.out.println("Grade: A");
                            else if (per >= 60) System.out.println("Grade: B");
                            else if (per >= 40) System.out.println("Grade: C");
                            else System.out.println("Grade: Fail");
                            break;
                        case 4:
                            System.out.print("Enter day number (1-7): ");
                            int day = sc.nextInt();
                            switch (day) {
                                case 1: System.out.println("Monday"); break;
                                case 2: System.out.println("Tuesday"); break;
                                case 3: System.out.println("Wednesday"); break;
                                case 4: System.out.println("Thursday"); break;
                                case 5: System.out.println("Friday"); break;
                                case 6: System.out.println("Saturday"); break;
                                case 7: System.out.println("Sunday"); break;
                                default: System.out.println("Invalid day!");
                            }
                            break;
                    }
                    break;
            }
            case 3:
                System.out.println("\n--- Looping Statements ---");
                System.out.println("1. Multiplication Table");
                System.out.println("2. Display numbers 1 to N");
                System.out.println("3. Sum of first N natural numbers");
                System.out.println("4. Fibonacci Series");
                int loopChoice = sc.nextInt();

                switch (loopChoice) {
                    case 1:
                        System.out.print("Enter number: ");
                        int n = sc.nextInt();
                        for (int i = 1; i <= n; i++)
                            System.out.println(n + " x " + i + " = " + (n * i));
                        break;
                }
            }
        } while (choice != 5);
    }
}
```

```

        case 2:
            System.out.print("Enter N: ");
            int N = sc.nextInt();
            for (int i = 1; i <= N; i++) System.out.print(i + " ");
            System.out.println();
            break;
        case 3:
            System.out.print("Enter N: ");
            int numN = sc.nextInt();
            int sum = 0;
            for (int i = 1; i <= numN; i++) sum += i;
            System.out.println("Sum = " + sum);
            break;
        case 4:
            System.out.print("Enter terms: ");
            int terms = sc.nextInt();
            int f1 = 0, f2 = 1;
            System.out.print("Fibonacci: " + f1 + " " + f2);
            for (int i = 3; i <= terms; i++) {
                int next = f1 + f2;
                System.out.print(" " + next);
                f1 = f2;
                f2 = next;
            }
            System.out.println();
            break;
    }
    break;

    case 4:
        System.out.println("\n--- Methods ---");
        System.out.println("1. Factorial");
        System.out.println("2. Prime Check");
        System.out.println("3. Max of Two Numbers");
        System.out.println("4. Area of Circle");
        int methodChoice = sc.nextInt();

        switch (methodChoice) {
            case 1:
                System.out.println("Enter number: ");
                int factNum = sc.nextInt();
                System.out.println("Factorial = " + factorial(factNum));
                break;
            case 2:
                System.out.println("Enter number: ");
                int primeNum = sc.nextInt();
                System.out.println(isPrime(primeNum) ? "Prime" : "Not Prime");
                break;
            case 3:
                System.out.print("Enter two numbers: ");
                int x = sc.nextInt(), y = sc.nextInt();
                System.out.println("Max = " + maxOfTwo(x, y));
                break;
            case 4:
                System.out.print("Enter radius: ");
                double r = sc.nextDouble();
                System.out.println("Area = " + areaOfCircle(r));
                break;
        }

        break;

    case 0:
        System.out.println("Exiting program...");
        break;

    default:
        System.out.println("Invalid choice!");
    }
} while (choice != 0);

sc.close();
}

public static long factorial(int n) {
    long fact = 1;
    for (int i = 1; i <= n; i++) {
        fact *= i;
    }
    return fact;
}

public static boolean isPrime(int n) {
    if (n <= 1)
        return false;

    for (int i = 2; i <= Math.sqrt(n); i++) {
        if (n % i == 0)
            return false;
    }
    return true;
}

public static int maxOfTwo(int a, int b) {
    return (a > b) ? a : b;
}

public static double areaOfCircle(double r) {
    return Math.PI * r * r;
}

```

}

6. OUTPUT / EXECUTION

Output 1 – Main Menu

```
studentUtility program
1.Student Information
2.Conditional Statements
3.Looping Statements
4.Utility methods
5.Exit
Enter your choice:
```

Output 2 – Student Information

```
Enter your choice:
1
Enter Name:
Ravi
Enter the Roll Number:
101
Enter Marks in 3 subjects
80 75 85

--- Student Result ---
Name: Ravi
Roll No: 101
Total Marks: 240
Percentage: 8000.0%
```

Output 3 – Conditional Statements

```
Conditional Statement
1. Even or Odd
2. Largest of Three Numbers
3. Grade based on Percentage
4. Day of the Week

Enter choice: 1
Enter a number: 24
Even

Enter choice: 2
Enter three numbers: 12 45 30
Largest: 45

Enter choice: 3
Enter percentage: 82
Grade: A

Enter choice: 4
Enter day number (1-7): 3
Wednesday
```

Output 4 – Looping Statements

```
--- Looping Statements ---
1. Multiplication Table
2. Display numbers 1 to N
3. Sum of first N natural numbers
4. Fibonacci Series

Enter choice: 1
Enter number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Enter choice: 2
Enter N: 5
1 2 3 4 5

Enter choice: 3
```

```
Enter N: 10
Sum = 55

Enter choice: 4
Enter terms: 7
Fibonacci: 0 1 1 2 3 5 8
```

Output 5 – Utility Methods

```
--- Methods ---
1. Factorial
2. Prime Check
3. Max of Two Numbers
4. Area of Circle

Enter choice: 1
Enter number:
5
Factorial = 120

Enter choice: 2
Enter number:
17
Prime

Enter choice: 3
Enter two numbers: 18 25
Max = 25

Enter choice: 4
Enter radius: 7
Area = 153.93804002589985
```

Output 6 – Exit

```
Enter your choice:
0
Exiting program...
```

7. RESULT

The Student Utility program was successfully implemented and its major operations were executed and verified. The outputs above correspond to the submitted source code.

8. LEARNING OUTCOME

Understood menu-driven programming, switch-case, if-else statements, loops, nested switch statements, Scanner input, and reusable user-defined methods in Java.